

Generador de 35 variables climaticos

Viacheslav Shalsiko

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Definir nombres de archivos raster

```
## Capas de datos fuente
prec_raster <- "CHELSA_12pr_1981-2010_V21_clip.tif"
tmin_raster <- "CHELSA_12tasmin_1981-2010_V21_clip.tif"
tmax_raster <- "CHELSA_12tasmax_1981-2010_V21_clip.tif"

## Capas de datos que se van a generar
tmean_raster <- "CHELSA_12tmean_1981-2010_V21_clip.tif"
biovars_raster <- "CHELSA_19biovars_1981-2010_V21_clip.tif"
solrad_raster <- "CHELSA_12solrad_1981-2010_V21_clip.tif"
pet_raster <- "CHELSA_12pet_1981-2010_V21_clip.tif"
envirem_raster <- "CHELSA_16envirem_1981-2010_V21_clip.tif"
```

Cargar bibliotecas

```
library(sp)
```

```
## Warning: package 'sp' was built under R version 4.5.2
```

```
library(raster)
```

```
## Warning: package 'raster' was built under R version 4.5.2
```

```
library(terra)
```

```
## terra 1.8.60
```

```
library(dismo)
```

```
## Warning: package 'dismo' was built under R version 4.5.2
```

```
library(envirem)
```

```
## Warning: package 'envirem' was built under R version 4.5.2
```

```
## Cargando paquete requerido: palinsol
```

```
## Warning: package 'palinsol' was built under R version 4.5.2
```

```
##  
## Please see the vignette at https://ptitle.github.io/envirem/ for a detailed walk-through of this package.
```

```
library(viridis)
```

```
## Warning: package 'viridis' was built under R version 4.5.2
```

```
## Cargando paquete requerido: viridisLite
```

```
gdal(lib = "all")
```

```
##      gdal      proj      geos  
## "3.11.0" "9.6.0" "3.13.1"
```

Cargar tres grupos de capas raster (precipitación, T mínima, T máxima) con el modulo raster

```
prec <- stack(prec_raster)  
names(prec) <- paste0("prec", seq(1, 12))  
prec
```

```
## class      : RasterStack  
## dimensions : 472, 585, 276120, 12  (nrow, ncol, ncell, nlayers)  
## resolution : 0.00833809, 0.008334101  (x, y)  
## extent      : -106.51, -101.6322, 17.98373, 21.91742  (xmin, xmax, ymin, ymax)  
## crs         : +proj=longlat +datum=WGS84 +no_defs  
## names       : prec1, prec2, prec3, prec4, prec5, prec6, prec7, prec8, prec9, prec10, prec11, prec12  
## min values  : 12.2,  4.5,  1.0,  0.2,  1.3, 26.4, 97.1, 87.7, 73.8, 27.6,  6.7,  3.4  
## max values  : 66.9, 28.4, 11.1, 17.0, 83.5, 397.2, 611.3, 563.4, 594.0, 215.5, 54.9, 34.0
```

```
tmin <- stack(tmin_raster)  
names(tmin) <- paste0("tmin", seq(1, 12))  
tmin
```

```
## class      : RasterStack  
## dimensions : 472, 585, 276120, 12  (nrow, ncol, ncell, nlayers)  
## resolution : 0.00833809, 0.008334101  (x, y)  
## extent      : -106.51, -101.6322, 17.98373, 21.91742  (xmin, xmax, ymin, ymax)  
## crs         : +proj=longlat +datum=WGS84 +no_defs  
## names       : tmin1, tmin2, tmin3, tmin4, tmin5, tmin6, tmin7, tmin8, tmin9, tmin10, tmin11, tmin12  
## min values  : -7.05, -6.65, -5.55, -3.85, -1.85, -0.05, -1.05, -1.15, -0.85, -1.95, -4.65, -6.55  
## max values  : 25.45, 24.95, 24.25, 24.55, 25.65, 26.55, 27.15, 27.15, 26.85, 27.05, 26.75, 26.05
```

```
tmax <- stack(tmax_raster)  
names(tmax) <- paste0("tmax", seq(1, 12))  
tmax
```

```
## class      : RasterStack
## dimensions : 472, 585, 276120, 12  (nrow, ncol, ncell, nlayers)
## resolution : 0.00833809, 0.008334101  (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742  (xmin, xmax, ymin, ymax)
## crs        : +proj=longlat +datum=WGS84 +no_defs
## names      : tmax1, tmax2, tmax3, tmax4, tmax5, tmax6, tmax7, tmax8, tmax9, tmax10, tmax11, tmax12
## min values :  6.45,  8.05, 10.15, 12.25, 12.95,  9.95,  7.95,  7.85,  7.55,  7.45,  7.15,  6.35
## max values : 33.25, 35.15, 37.45, 39.85, 40.15, 36.65, 33.85, 33.95, 33.45, 33.55, 33.75, 33.25
```

Guardar una plantilla de capa raster

```
writeRaster(prec[[1]],
            filename="raster_template_clip.tif",
            format="GTiff", datatype="FLT4S", overwrite=TRUE)
```

Generar y guardar 12 variables de temperatura media mensual

```
tmean <- (tmin + tmax) / 2
names(tmean) <- paste0("tmean", seq(1, 12))
tmean
```

```
## class      : RasterBrick
## dimensions : 472, 585, 276120, 12  (nrow, ncol, ncell, nlayers)
## resolution : 0.00833809, 0.008334101  (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742  (xmin, xmax, ymin, ymax)
## crs        : +proj=longlat +datum=WGS84 +no_defs
## source     : memory
## names      : tmean1, tmean2, tmean3, tmean4, tmean5, tmean6, tmean7, tmean8, tmean9, tmean10, tmean11, tmean12
## min values : -0.3000002, 0.7000000, 2.2999997, 4.2000000, 5.5499999, 4.9499999, 3.4499999, 3.3500001, 2.7499999, 1.2500000, -0.1000001
## max values : 26.80, 27.55, 28.80, 31.10, 32.20, 30.50, 29.45, 29.40, 28.45, 28.35, 28.20, 27.15
```

```
writeRaster(tmean,
            filename=tmean_raster,
            format="GTiff", datatype="FLT4S", overwrite=TRUE)
```

Generar y guardar 19 variables bioclimaticas con el modulo dismo

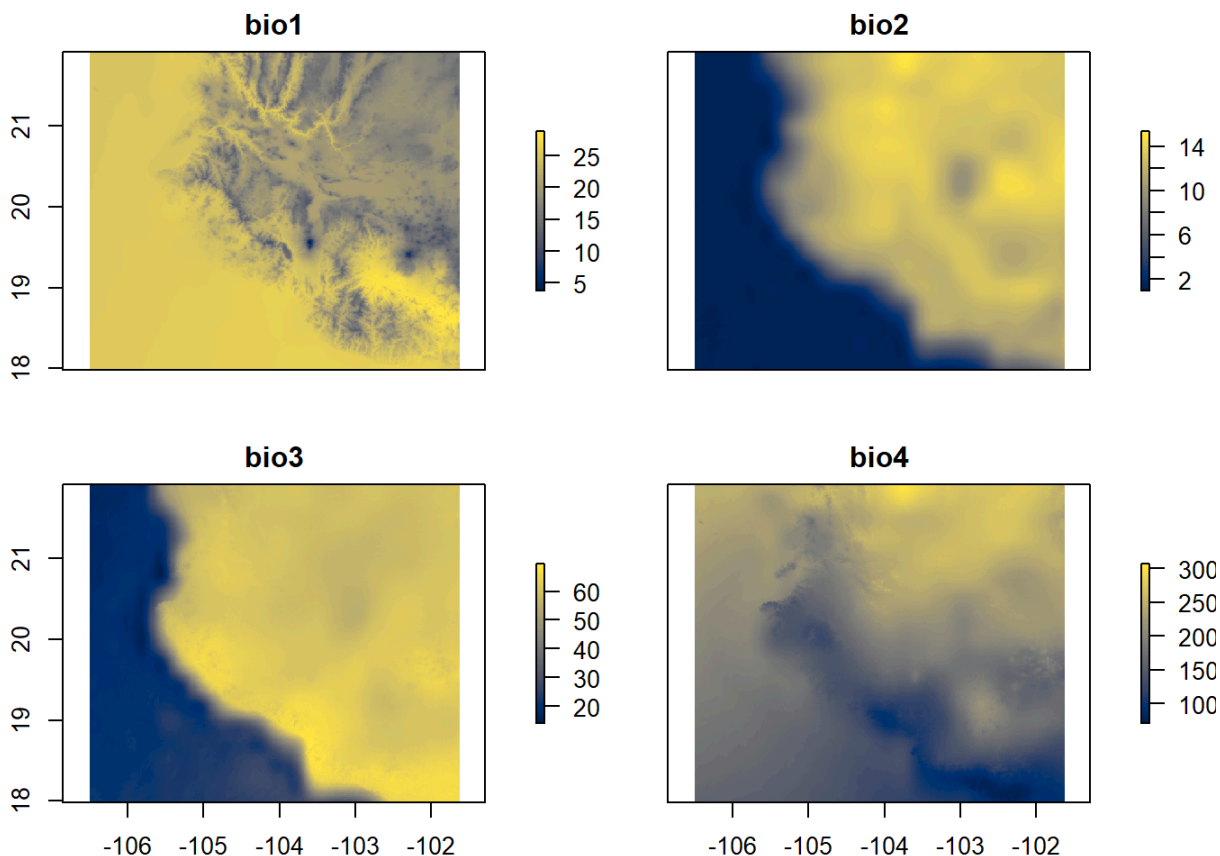
```
biovars <- biovars(prec, tmin, tmax)
biovars
```

```
## class      : RasterBrick
## dimensions : 472, 585, 276120, 19  (nrow, ncol, ncell, nlayers)
## resolution : 0.00833809, 0.008334101  (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742  (xmin, xmax, ymin, ymax)
## crs        : +proj=longlat +datum=WGS84 +no_defs
## source     : memory
## names      :      bio1,      bio2,      bio3,      bio4,      bio5,      bio6,      bio7,
bio8,      bio9,      bio10,      bio11,      bio12,      bio13,      bio14,      bio15, ...
## min values :  2.6208333,  0.9749997, 14.2512053, 71.8307750, 12.9499998, -7.0500002,  4.5000000,
3.3833333,  2.3999999,  4.8999999,  0.0999999, 471.2000034, 101.6999969,  0.2000000, 86.4673568, ...
## max values : 28.78750, 15.34167, 69.44445, 306.40234, 40.15000, 24.25000, 25.10000,
29.10000, 28.91667, 31.21667, 27.13333, 2521.90005, 611.29999, 11.10000, 131.15292, ...
```

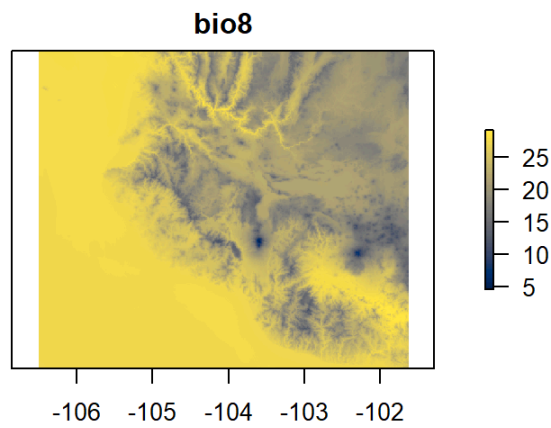
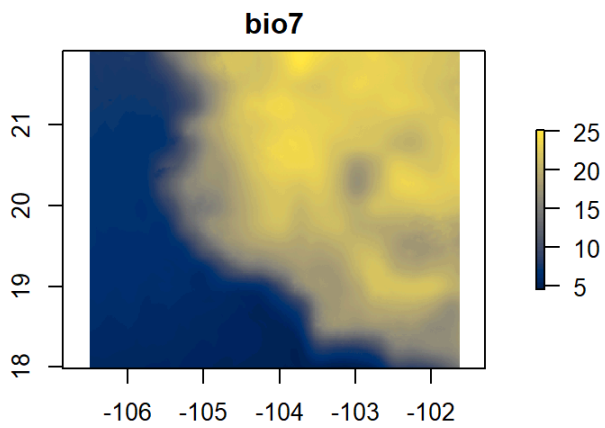
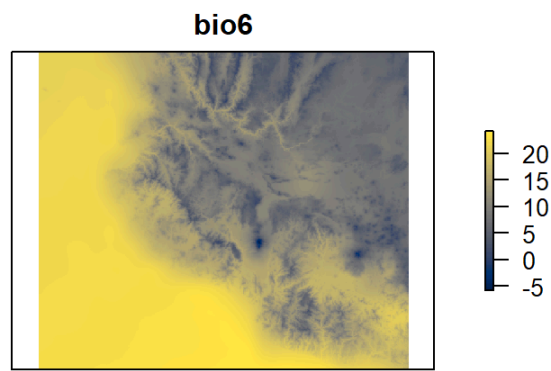
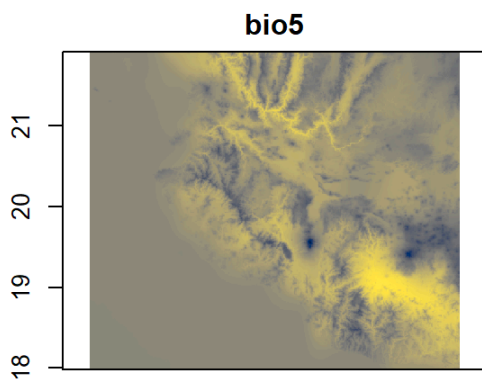
```
writeRaster(biovars,
            filename=biovars_raster,
            format="GTiff", datatype="FLT4S", overwrite=TRUE)
```

Visualizar variables bioclim

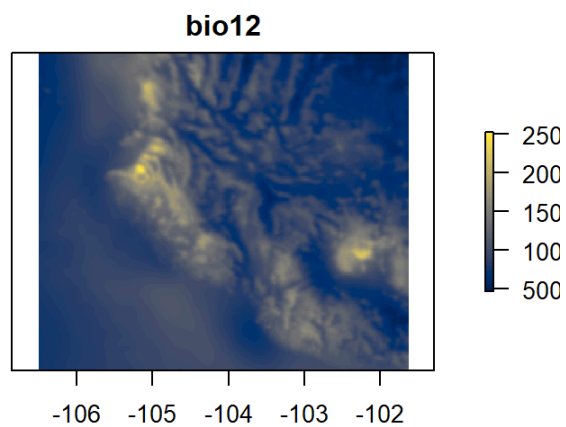
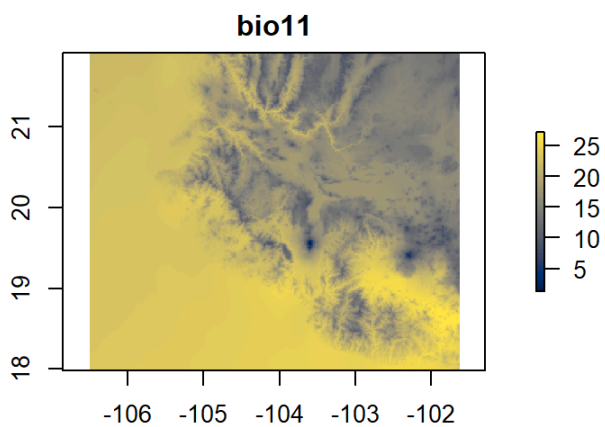
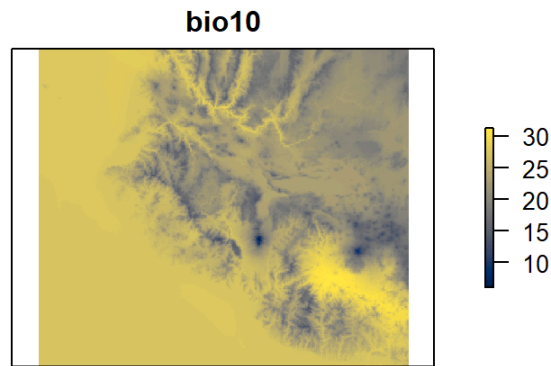
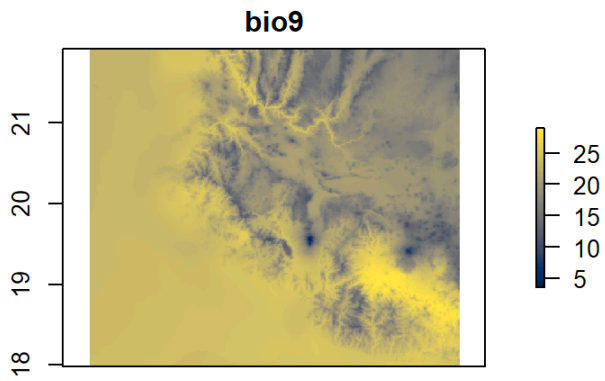
```
plot(biovars[[1:4]], col = cividis(100))
```



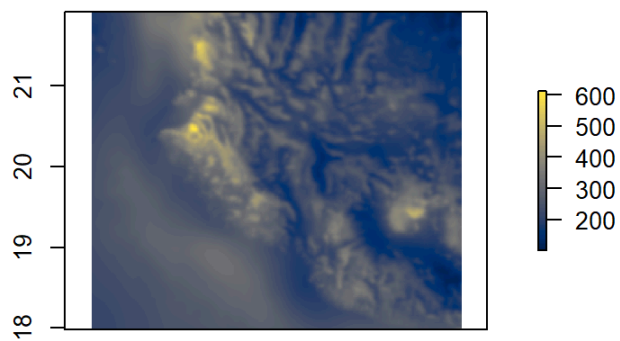
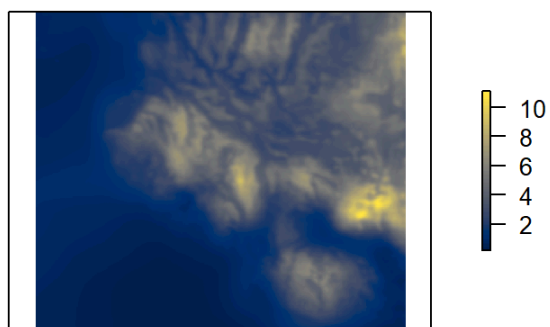
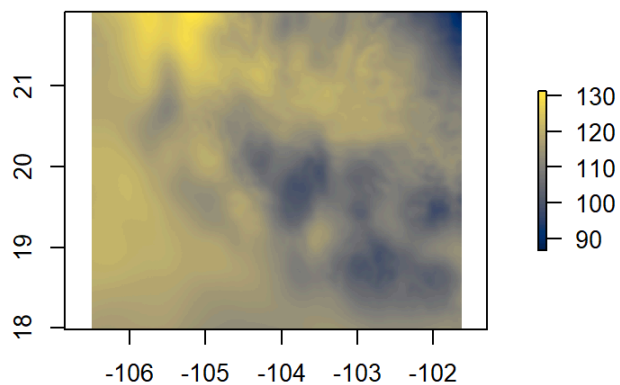
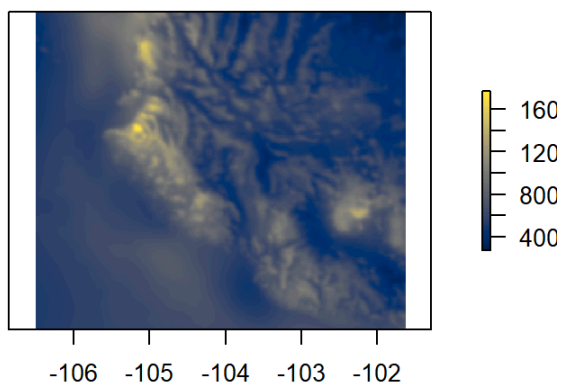
```
plot(biovars[[5:8]], col = cividis(100))
```



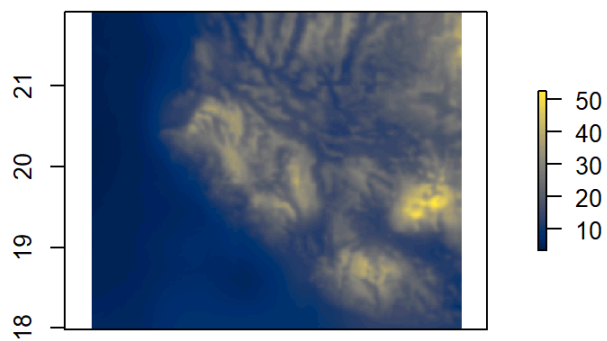
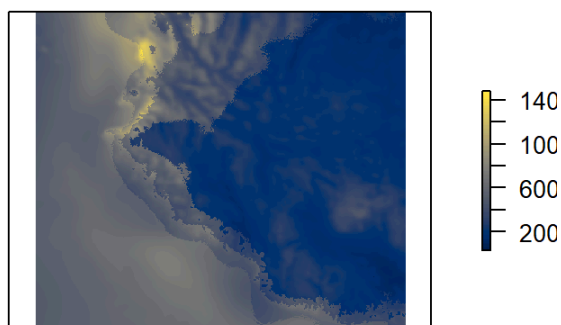
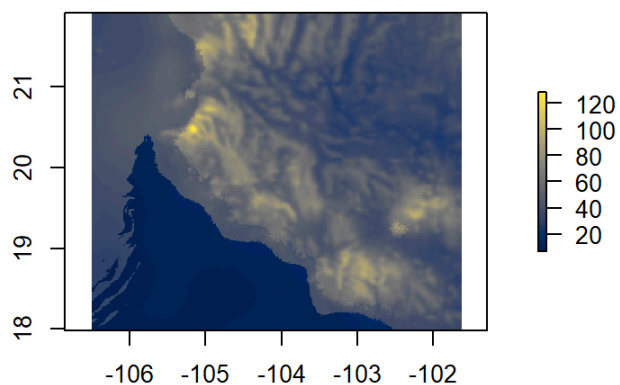
```
plot(biovars[[9:12]], col = cividis(100))
```



```
plot(biovars[[13:16]], col = cividis(100))
```

bio13**bio14****bio15****bio16**

```
plot(biovars[[17:19]], col = cividis(100))
```

bio17**bio18****bio19**

Cargar plantilla de raster previamente cargada con el modulo terra

```
raster_template <- terra::rast("raster_template_clip.tif")
raster_template
```

```
## class      : SpatRaster
## size       : 472, 585, 1  (nrow, ncol, nlyr)
## resolution : 0.00833809, 0.008334101  (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742  (xmin, xmax, ymin, ymax)
## coord. ref. : lon/lat WGS 84 (EPSG:4326)
## source     : raster_template_clip.tif
## name       : raster_template_clip
## min value   :                12.2
## max value   :                66.9
```

Redefinir las funciones para generar las capas de insolación

```
ETsolradRastersMy <- function(rasterTemplate, year, outputDir = NULL, ...) {

  if (terra::is.lonlat(rasterTemplate)) {
    isLongLat <- TRUE
  } else {
    isLongLat <- FALSE
  }

  if (!isLongLat) {
    rasterTemplate2 <- rasterTemplate
    rasterTemplate <- terra::project(rasterTemplate, 'EPSG:4326')
  }

  # extract latitudes from cells
  latvals <- terra::yFromRow(rasterTemplate, row = 1:nrow(rasterTemplate))

  # for each month, calculate insolation and fill raster
  solradStack <- terra::rast(rasterTemplate, nlyrs = 12)
  for (i in 1:12) {

    message(i, ' ', appendLF = FALSE)
    RA <- sapply(latvals, function(x) calcSolRadMy(year = year, lat = x, month = i))
    tmp <- rep(RA, each = terra::ncol(rasterTemplate))
    ## siguiente linea se corrige sobre el código original de ENVIREM, para generar capas correctas
    tmp <- matrix(data = tmp, nrow = terra::nrow(rasterTemplate), ncol = terra::ncol(rasterTemplate), by
row = TRUE)
    ras <- terra::init(rasterTemplate, fun = tmp)
    solradStack[[i]] <- ras
  }

  names(solradStack) <- paste0("et_solrad_", sprintf("%02d", 1:12))

  if (!isLongLat) {
    # project to input projection
    solradStack <- terra::project(solradStack, rasterTemplate2)
    # mask NA regions
    solradStack <- terra::mask(solradStack, rasterTemplate2)
  } else {
    # mask NA regions
    solradStack <- terra::mask(solradStack, rasterTemplate)
  }

  return(solradStack)
}

calcSolRadMy <- function(year, lat, month) {
  # year is number of years after 1950
  # lat is in degrees
  # month is numeric, will be converted into 15th day of the month to get mid-month mean solar radiation

  # define orbit
  orb <- la04(t = year)

  # convert date to julian day
  day <- c(15, 46, 74, 105, 135, 166, 196, 227, 258, 288, 319, 349)[month]
```



```

# convert julian day to day based on 360-day year
day <- day * 360/365

# calculate solar longitude for given year
# t = time after 1950, day is in 360-day calendar
solarLong <- day2l(orb, day = day)

# returns daily mean insolation in Watts/m2
solrad <- Insol(orbit = orb, long = solarLong, lat = lat*pi/180)

# convert to MJ/m2 by multiplying by 0.0864, and then by 0.408 to get equivalent evaporation in mm / day
solrad * 0.0864 * 0.408
}

```

Generar las 12 capas de radiación solar

```

## el raster de plantilla tiene que ser en el formato
## SpatRaster del paquete terra
## parámetro year = 50 corresponde al año 2000
solrad <- ETSolradRastersMy(raster_template, year = 50, outputDir = NULL)

```

```
## 1 2 3 4 5 6 7 8 9 10 11 12
```

```
solrad
```

```

## class      : SpatRaster
## size       : 472, 585, 12  (nrow, ncol, nlyr)
## resolution : 0.00833809, 0.008334101  (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742  (xmin, xmax, ymin, ymax)
## coord. ref.: lon/lat WGS 84 (EPSG:4326)
## source(s)  : memory
## varname    : raster_template_clip
## names      : et_solrad_01, et_solrad_02, et_solrad_03, et_solrad_04, et_solrad_05, et_solrad_06, ...
## min values :    10.45229,    12.13591,    13.92042,    15.40023,    15.91509,    15.94529, ...
## max values :    11.32504,    12.83656,    14.35822,    15.49674,    16.10268,    16.28163, ...

```

Guardar las capas de radiación solar

```

writeRaster(solrad,
            filename=solrad_raster,
            datatype="FLT4S",
            NAflag=-9999,
            overwrite=TRUE)

```

Cargar capas raster previamente guardadas con el modulo terra

```

t_prec <- terra::rast(prec_raster)
names(t_prec) <- paste0("prec", sprintf("%02d", 1:12))
t_prec

```

```
## class      : SpatRaster
## size       : 472, 585, 12 (nrow, ncol, nlyr)
## resolution : 0.00833809, 0.008334101 (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742 (xmin, xmax, ymin, ymax)
## coord. ref. : lon/lat WGS 84 (EPSG:4326)
## source      : CHELSA_12pr_1981-2010_V21_clip.tif
## names       : prec01, prec02, prec03, prec04, prec05, prec06, ...
## min values  : 12.2, 4.5, 1.0, 0.2, 1.3, 26.4, ...
## max values  : 66.9, 28.4, 11.1, 17.0, 83.5, 397.2, ...
```

```
t_tmin <- terra::rast(tmin_raster)
names(t_tmin) <- paste0("tmin", sprintf("%02d", 1:12))
t_tmin
```

```
## class      : SpatRaster
## size       : 472, 585, 12 (nrow, ncol, nlyr)
## resolution : 0.00833809, 0.008334101 (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742 (xmin, xmax, ymin, ymax)
## coord. ref. : lon/lat WGS 84 (EPSG:4326)
## source      : CHELSA_12tasmin_1981-2010_V21_clip.tif
## names       : tmin01, tmin02, tmin03, tmin04, tmin05, tmin06, ...
## min values  : -7.05, -6.65, -5.55, -3.85, -1.85, -0.05, ...
## max values  : 25.45, 24.95, 24.25, 24.55, 25.65, 26.55, ...
```

```
t_tmax <- terra::rast(tmax_raster)
names(t_tmax) <- paste0("tmax", sprintf("%02d", 1:12))
t_tmax
```

```
## class      : SpatRaster
## size       : 472, 585, 12 (nrow, ncol, nlyr)
## resolution : 0.00833809, 0.008334101 (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742 (xmin, xmax, ymin, ymax)
## coord. ref. : lon/lat WGS 84 (EPSG:4326)
## source      : CHELSA_12tasmax_1981-2010_V21_clip.tif
## names       : tmax01, tmax02, tmax03, tmax04, tmax05, tmax06, ...
## min values  : 6.45, 8.05, 10.15, 12.25, 12.95, 9.95, ...
## max values  : 33.25, 35.15, 37.45, 39.85, 40.15, 36.65, ...
```

```
t_tmean <- terra::rast(tmean_raster)
names(t_tmean) <- paste0("tmean", sprintf("%02d", 1:12))
t_tmean
```

```
## class      : SpatRaster
## size       : 472, 585, 12 (nrow, ncol, nlyr)
## resolution : 0.00833809, 0.008334101 (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742 (xmin, xmax, ymin, ymax)
## coord. ref. : lon/lat WGS 84 (EPSG:4326)
## source      : CHELSA_12tmean_1981-2010_V21_clip.tif
## names       : tmean01, tmean02, tmean03, tmean04, tmean05, tmean06, ...
## min values  : -0.3000002, 0.70, 2.3, 4.2, 5.55, 4.95, ...
## max values  : 26.7999992, 27.55, 28.8, 31.1, 32.20, 30.50, ...
```

```
t_biovars <- terra::rast(biovars_raster)
names(t_biovars) <- paste0("biovars", sprintf("%02d", 1:19))
t_biovars
```

```
## class      : SpatRaster
## size       : 472, 585, 19  (nrow, ncol, nlyr)
## resolution : 0.00833809, 0.008334101  (x, y)
## extent     : -106.51, -101.6322, 17.98373, 21.91742  (xmin, xmax, ymin, ymax)
## coord. ref.: lon/lat WGS 84 (EPSG:4326)
## source     : CHELSA_19biovars_1981-2010_V21_clip.tif
## names      : biovars01, biovars02, biovars03, biovars04, biovars05, biovars06, ...
## min values : 2.620833, 0.9749997, 14.25121, 71.83077, 12.95, -7.05, ...
## max values : 28.787500, 15.3416672, 69.44445, 306.40234, 40.15, 24.25, ...
```

Establecer esquemas de nombres de capas requeridos por las funciones del modulo envirem

```
assignNames(precip = 'prec##',
            tmin = 'tmin##',
            tmax = 'tmax##',
            tmean = 'tmean##',
            solrad = 'et_solrad_##',
            pet = 'PET_##')

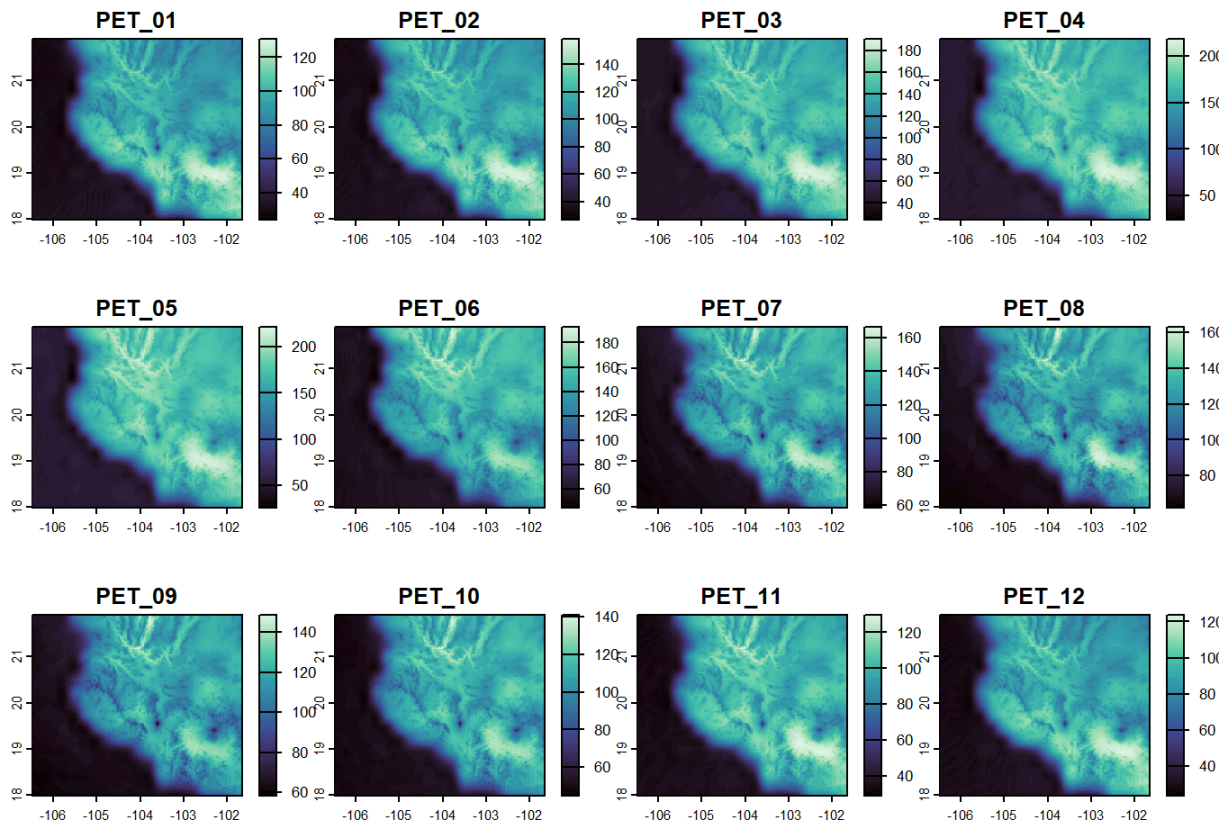
allvar <- c("annualPET", "aridityIndexThornthwaite", "climaticMoistureIndex", "continentality", "embergerQ",
"growingDegDays0", "growingDegDays5", "maxTempColdest", "minTempWarmest", "meanTempColdest", "meanTempWarmest",
"monthCountByTemp10", "PETColdestQuarter", "PETDriestQuarter", "PETseasonality", "PETWarmestQuarter", "PETWettestQuarter", "thermicityIndex")
```

Generar el raster stack en blanco para los resultados

```
reslist <- vector('list', length = length(allvar))
names(reslist) <- allvar
```

Generar PET mensual con el modulo envirem

```
tempRange <- abs(t_tmax - t_tmin)
monthlyPET <- monthlyPET(t_tmean, solrad, tempRange, tempScale = 1)
plot(monthlyPET, col = mako(100))
```



Guardar las capas de PET mensual

```
writeRaster(monthlyPET,
            filename=pet_raster,
            datatype = "FLT4S",
            NAflag = -9999,
            overwrite=TRUE)
```

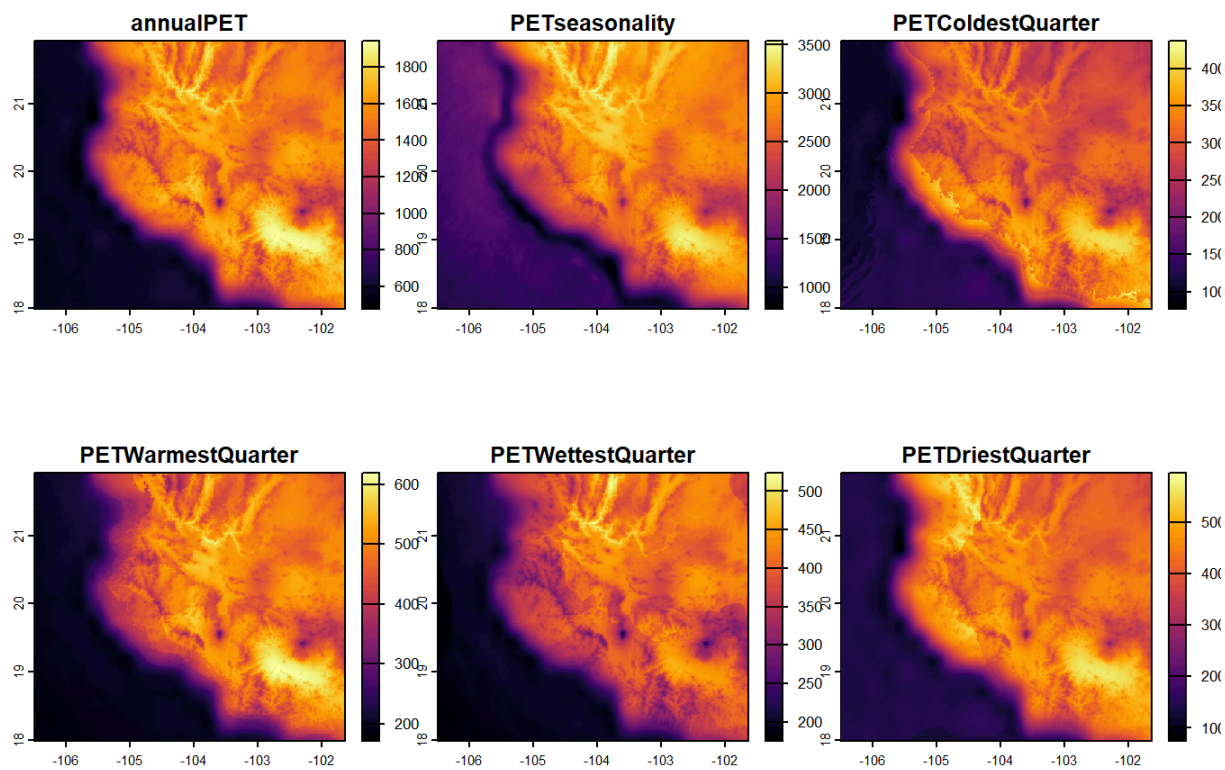
Generar PET derivados con el modulo envirem

```
annualPET <- sum(monthlyPET)
names(annualPET) <- "annualPET"
reslist[['annualPET']] <- annualPET

PETseas <- PETseasonality(monthlyPET)
reslist[['PETseasonality']] <- PETseas

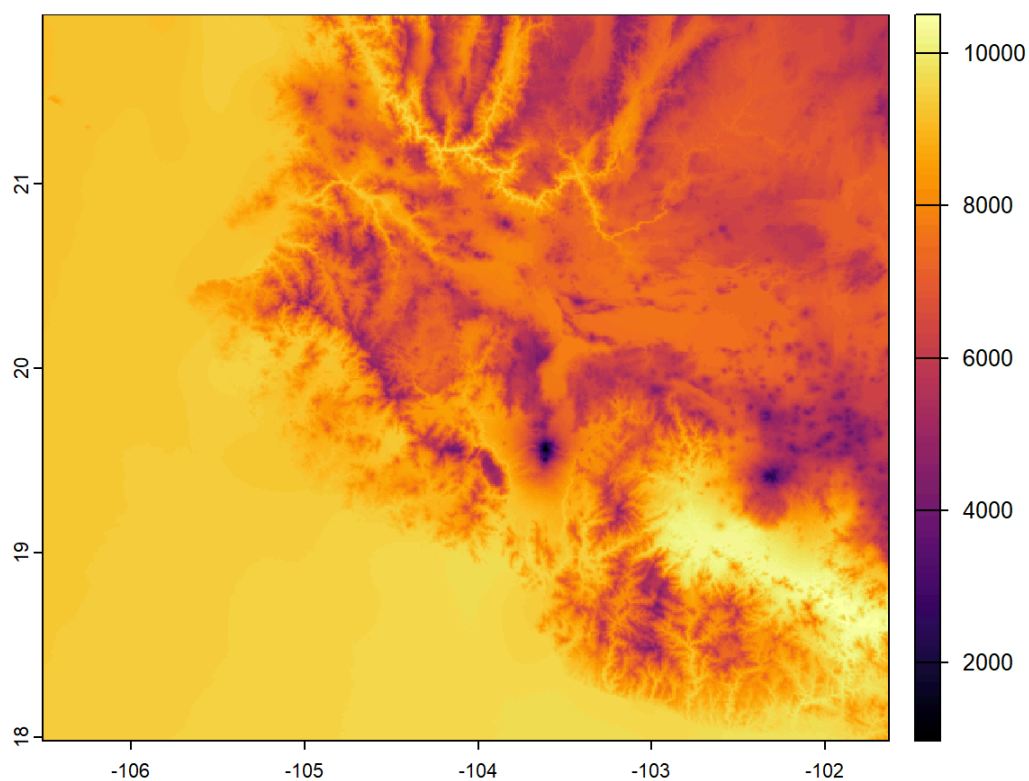
PETextr <- petExtremes(monthlyPET, t_prec, t_tmean)
reslist[['PETColdestQuarter']] <- PETextr[[1]]
reslist[['PETWarmestQuarter']] <- PETextr[[2]]
reslist[['PETWettestQuarter']] <- PETextr[[3]]
reslist[['PETDriestQuarter']] <- PETextr[[4]]

fullPET <- c(annualPET, PETseas, PETextr[[1]], PETextr[[2]], PETextr[[3]], PETextr[[4]])
plot(fullPET, col = inferno(100))
```



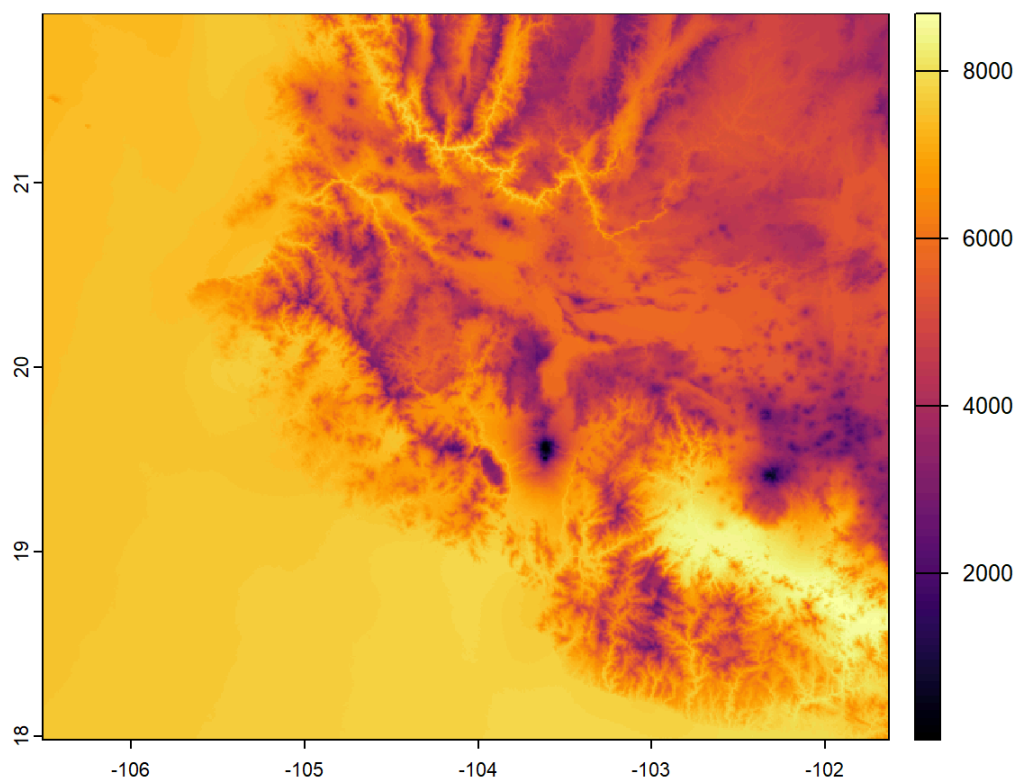
Generar otros variables con el modulo envirem

```
gdd0 <- growingDegDays(t_tmean, baseTemp = 0)
plot(gdd0, col = inferno(100))
```



```
reslist[['growingDegDays0']] <- gdd0

gdd5 <- growingDegDays(t_tmean, baseTemp = 5)
plot(gdd5, col = inferno(100))
```

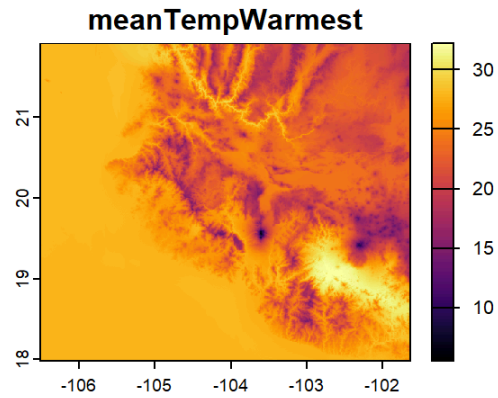
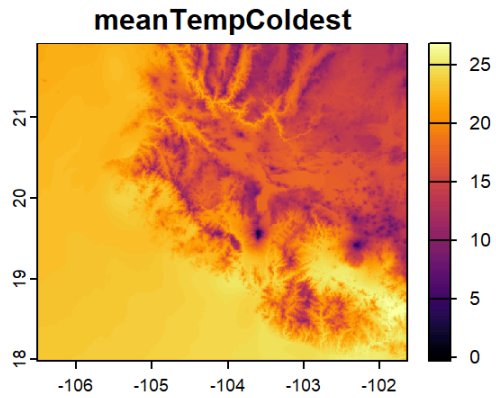
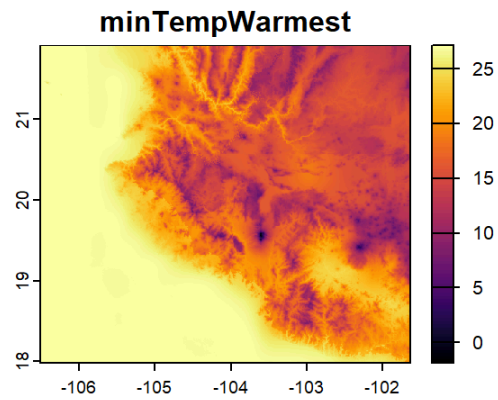
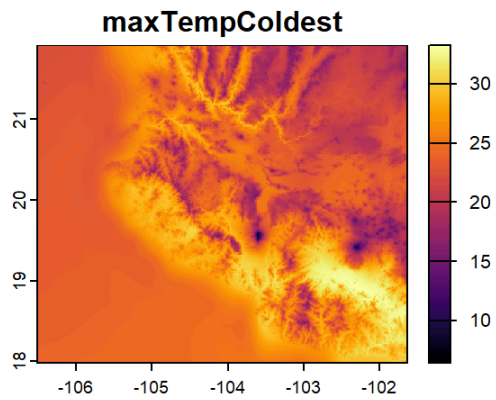


```
reslist[['growingDegDays5']] <- gdd5
```

```
tempExtremes <- otherTempExtremes(t_tmean, t_tmin, t_tmax)
```

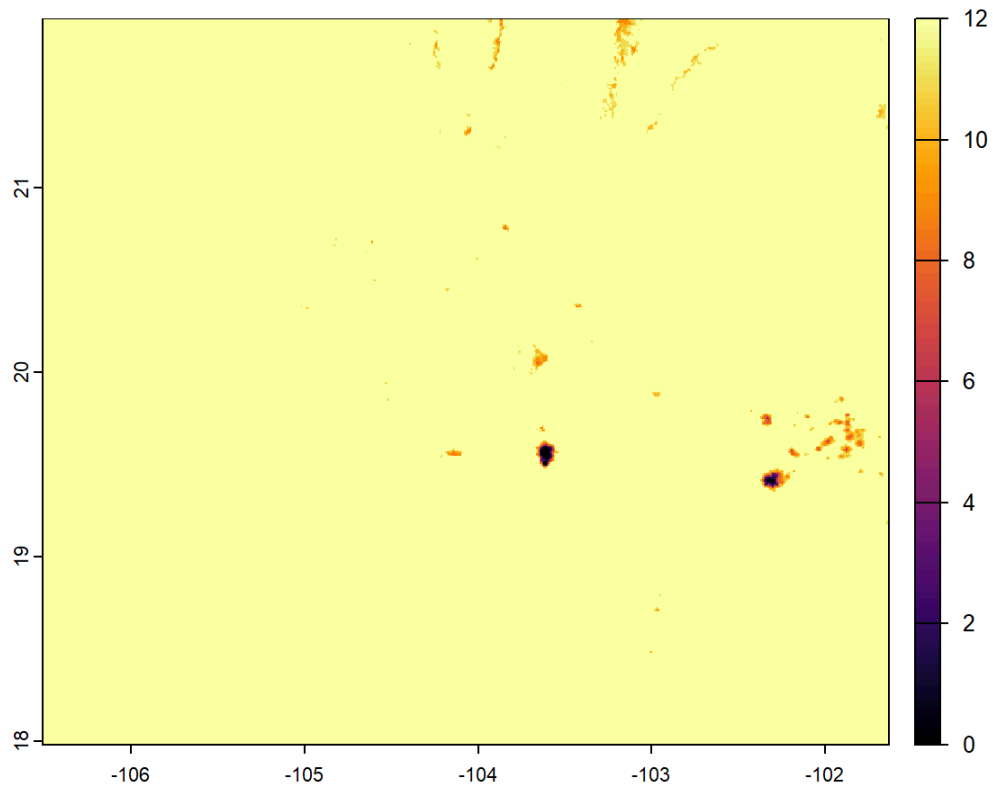
```
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
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## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
## Warning: [extract] nothing to extract
```

```
plot(tempExtremes, col = inferno(100))
```



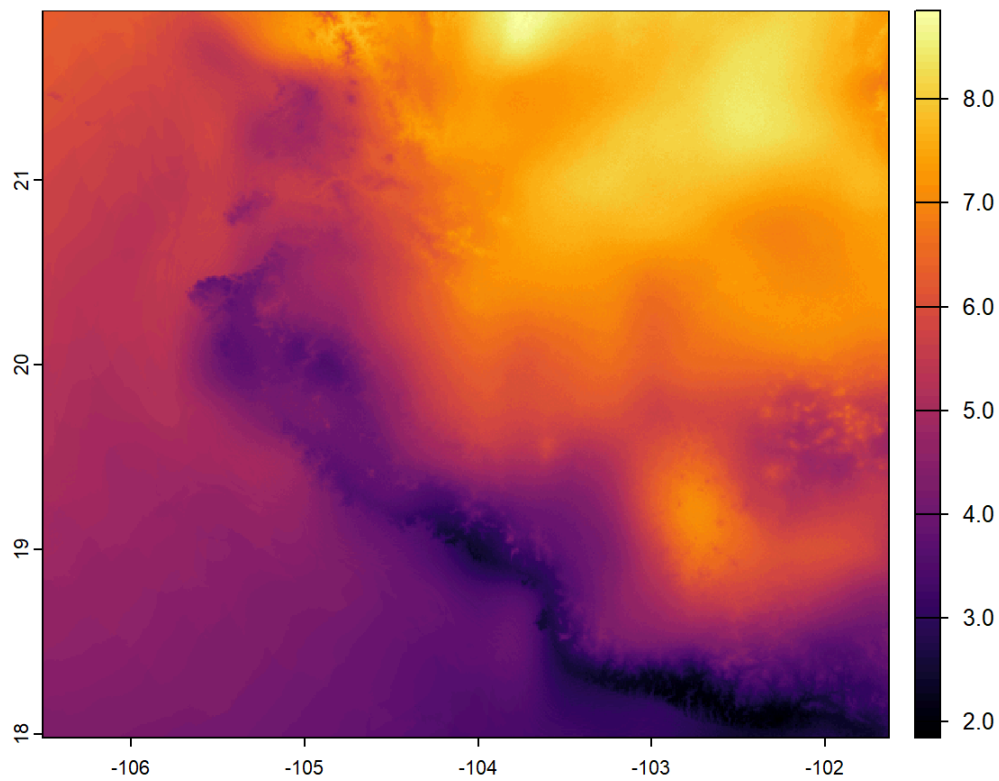
```
reslist[['minTempWarmest']] <- tempExtremes[['minTempWarmest']]  
reslist[['maxTempColdest']] <- tempExtremes[['maxTempColdest']]  
reslist[['meanTempWarmest']] <- tempExtremes[['meanTempWarmest']]  
reslist[['meanTempColdest']] <- tempExtremes[['meanTempColdest']]
```

```
monthCount10deg <- monthCountByTemp(t_tmean, minTemp = 10)  
plot(monthCount10deg, col = inferno(100))
```

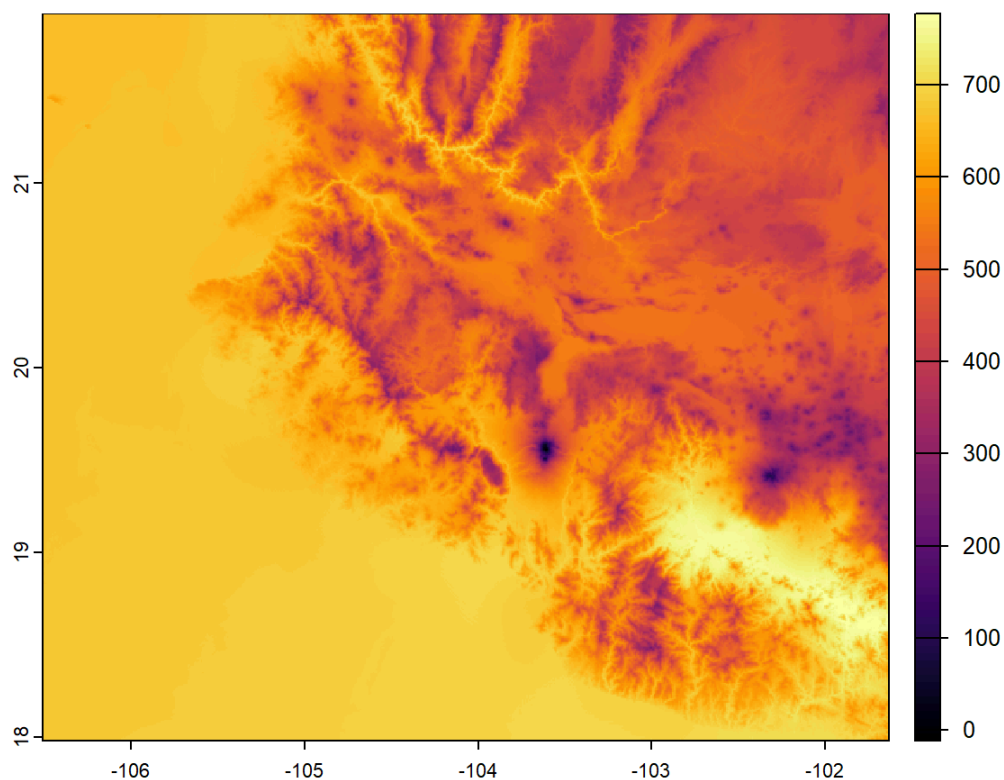
```
reslist[['monthCountByTemp10']] <- monthCount10deg
```

```
ci <- continentality(tmax = tempExtremes[[4]], tmin = tempExtremes[[3]])
plot(ci, col = inferno(100))
```



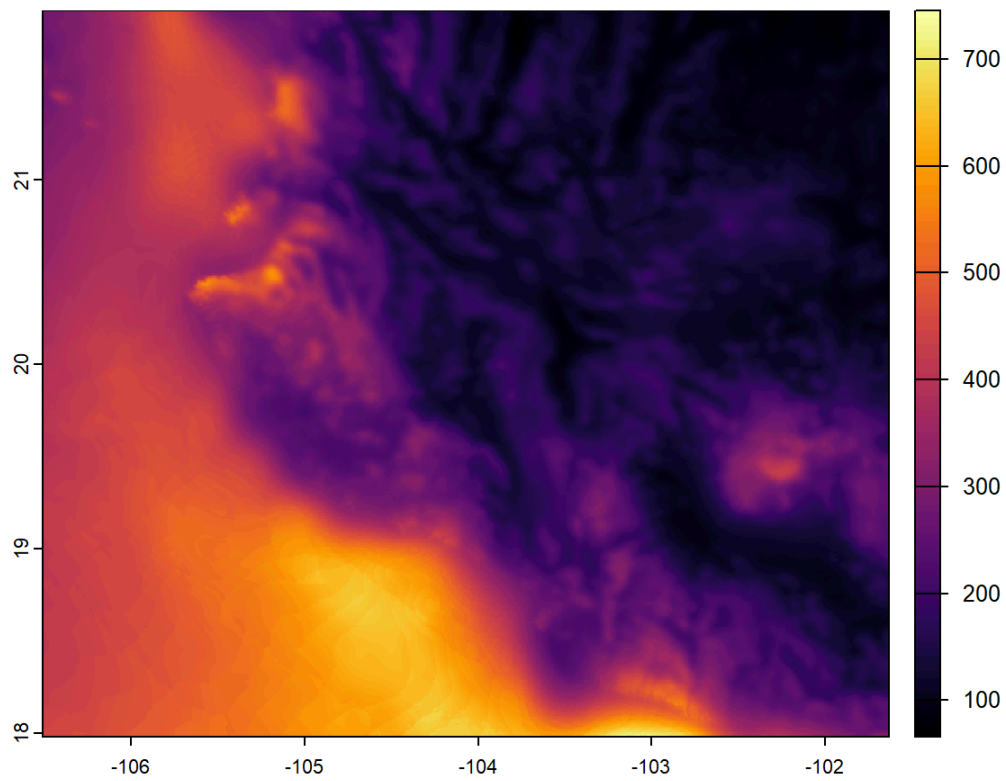

```
reslist[['continentiality']] <- ci
```

```
thermInd <- thermicityIndex(annualTemp = t_biovars[[1]], minTemp = t_biovars[[6]], maxTemp = tempExtremes  
[[1]], continentiality = ci)  
plot(thermInd, col = inferno(100))
```



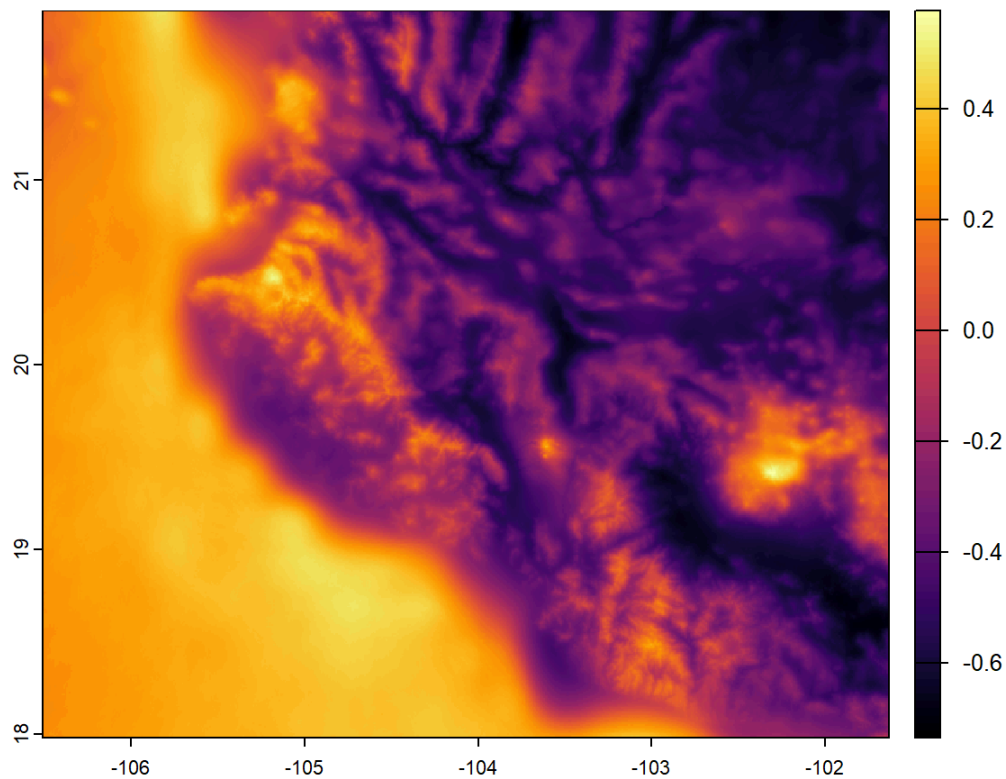
```
reslist[['thermicityIndex']] <- thermInd
```

```
emberger <- embergerQ(t_biovars[[12]], t_biovars[[5]], t_biovars[[6]])  
plot(emberger, col = inferno(100))
```



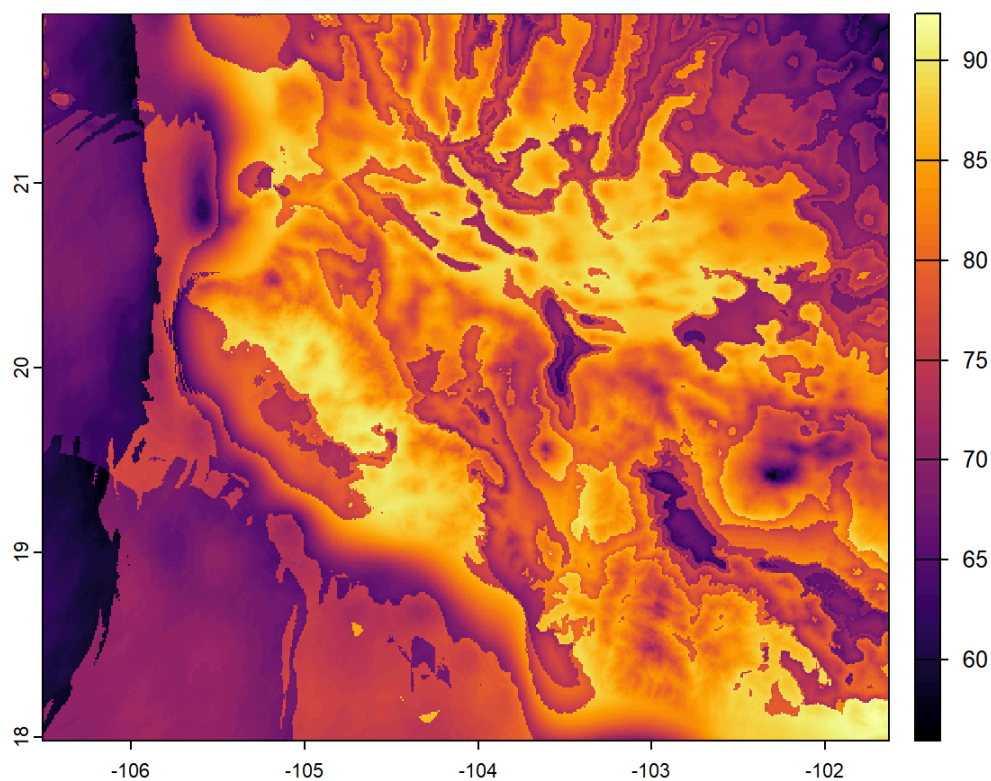
```
reslist[['embergerQ']] <- emberger
```

```
cmi <- climaticMoistureIndex(t_biovars[[12]], annualPET)
plot(cmi, col = inferno(100))
```



```
reslist[['climaticMoistureIndex']] <- cmi
```

```
aridIndThorn <- aridityIndexThornthwaite(t_prec, monthlyPET)  
plot(aridIndThorn, col = inferno(100))
```



```
reslist[['aridityIndexThornthwaite']] <- aridIndThorn
```

Guardar capas envirem generadas

```
reslist <- terra::rast(reslist)  
reslist <- reslist[[allvar]]  
  
writeRaster(reslist,  
            filename=envirem_raster,  
            datatype = "FLT4S",  
            NAflag = -9999,  
            overwrite=TRUE)
```